



BITS Pilani
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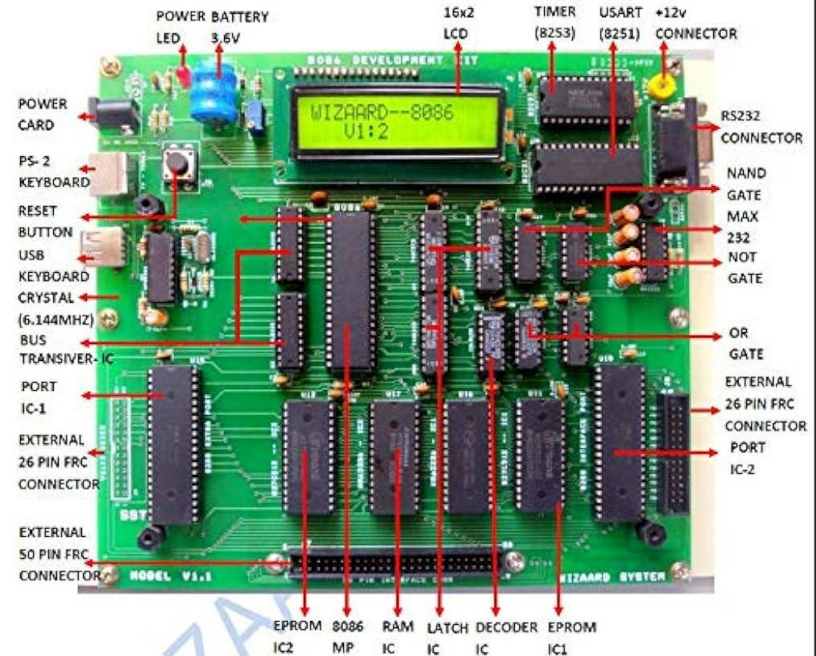
8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

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8086



WIZAARD

8086 and 8088 Device Specifications

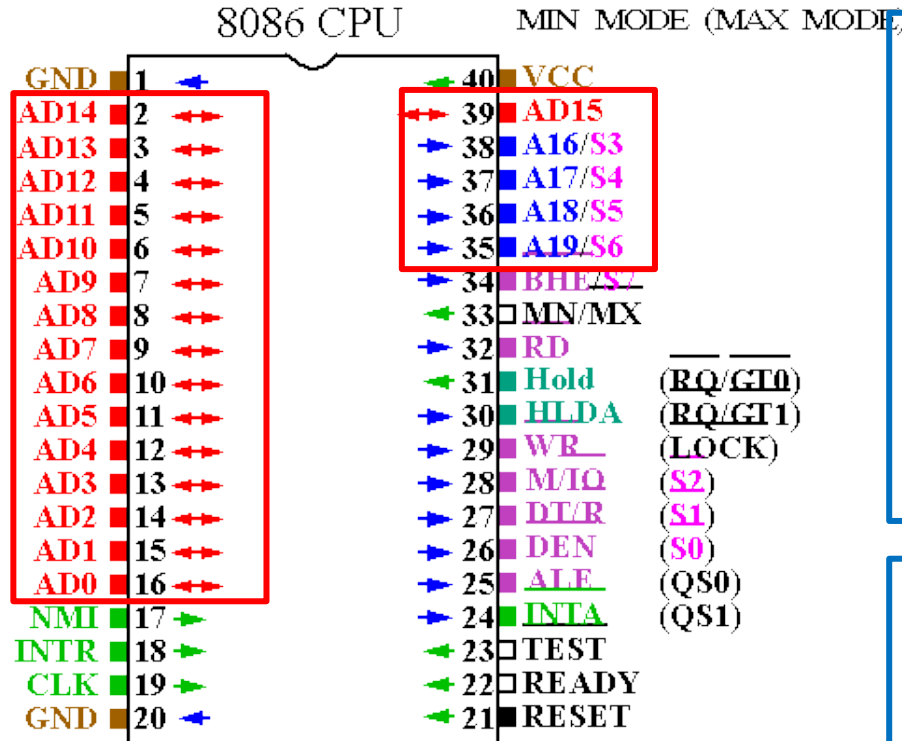
Both are packaged in DIP (Dual In-Line Packages).

- 8086: 16-bit microprocessor with a 16-bit data bus
- 8088: 16-bit microprocessor with an 8-bit data bus.

Both are 5V parts:

- 8086: Draws a maximum supply current of 360mA.
- 8088: Draws a maximum supply current of 340mA.
- 80C86/80C88: CMOS version draws 10mA with temp spec -40 to 225degF.

8086/8088 Pin Out



AD0-AD15(Bidirectional) Address/Data bus :
Low order address bus; these are multiplexed with data.

When AD lines are used to transmit memory address the symbol A is used instead of AD, for example A0-A15.

When AD lines are used to transmit data address the symbol D is used instead of AD, for example D0-D15.

A16/S3, A17/S4, A18/S5, A19/S6
High order address bus. These are multiplexed with Status Signals.

Why Intel multiplexed the data and address bus?

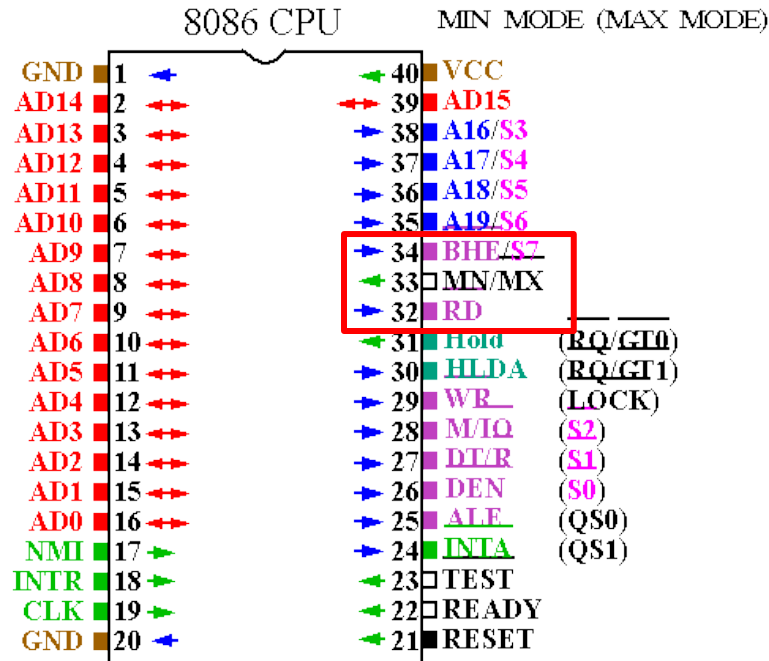
Reduce the pin count

Fit the chip in specific package.

- **Time multiplexing:** Use different time to send address and data.
- **Demultiplexing:** A latch is used to demultiplex the address and data bus.
- **Performance impact:** One extra memory cycle is used to send the address.



8086/8088 Pin Out



$\overline{\text{BHE}}$ (Active Low)/S7(Output) Bus High Enable/Status

It is used to enable data on to the most significant half of data bus, D8-D15. 8-bit device connected to upper half of the data bus use BHE (Active Low) signal. It is multiplexed with status signal S7 (always 1).

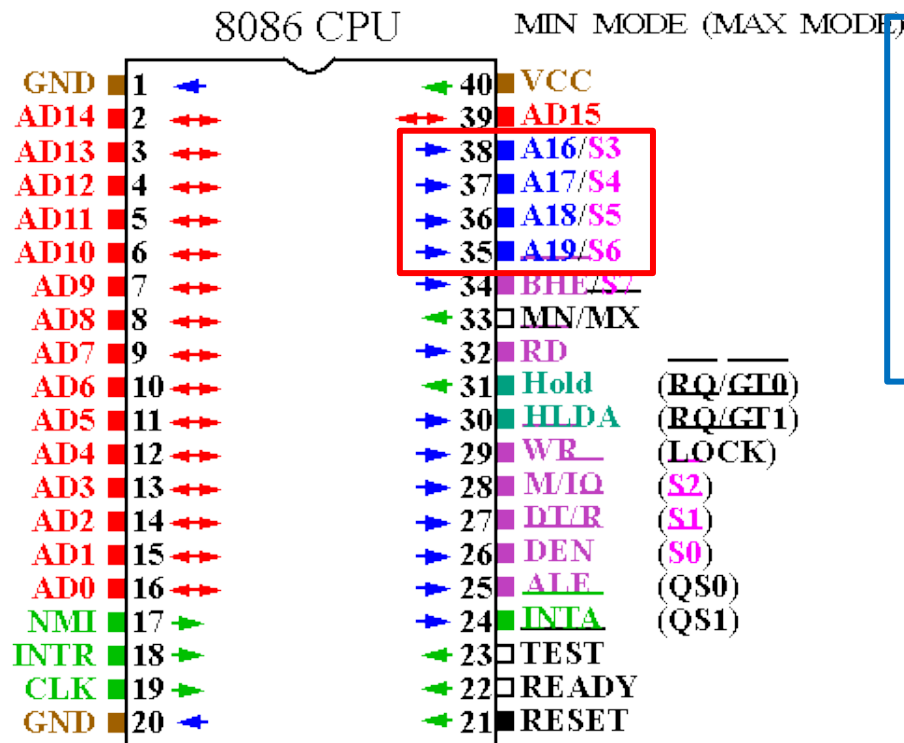
MN/ $\overline{\text{MX}}$ High Enable/Status

This pin indicates the mode of the processor to operate on.

$\overline{\text{RD}}$ (Read) (Active Low)

The signal is used for read operation. It is an output signal. It is active when low.

8086/8088 Pin Out

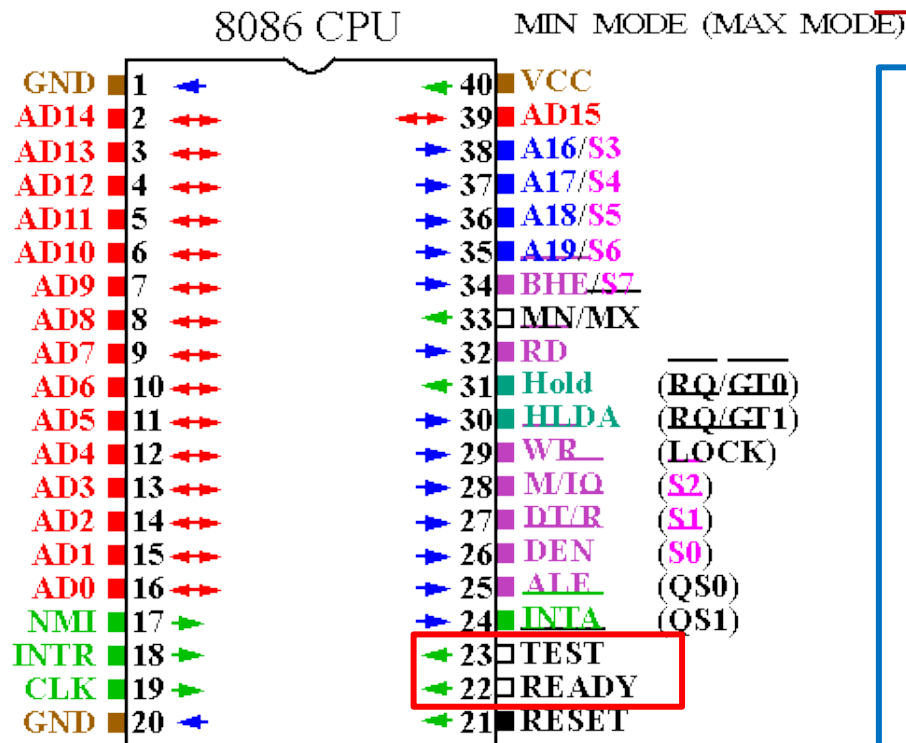


S3, S4, S5, S6 (Status Signals) : S6 is always logic 0
S5 indicates the condition of the IF flag bit.

S4 and S3 show which segment is access during the current bus cycle. These two status bits could be Used to address four separate 1M byte memory Bank by decoding them as A_{21} and A_{20} .

S_4	S_3	Function
0	0	Extra segment
0	1	Stack segment
1	0	Code or no segment
1	1	Data segment

8086/8088 Pin Out



Test

TEST input is tested by the 'WAIT' instruction. 8086 will enter a **wait state** after execution of the **WAIT instruction** and will resume execution only when the **TEST** is made low by an active hardware.

If **Test = 0** Wait, If **Test = 1** 8086 resumes execution

This is used to synchronize an external activity to the processor internal operation.

Ready

This is used to **insert the Wait states** into the timing of the Microprocessor. If this **signal is low** the Microprocessor will enter into **wait states** and remains idle.

8086/8088 Pin Out

8086 CPU				MIN MODE (MAX MODE)			
GND	1	←		40	VCC		
AD14	2	↔		39	AD15	↔	
AD13	3	↔		38	A16/S3	←	
AD12	4	↔		37	A17/S4	←	
AD11	5	↔		36	A18/S5	←	
AD10	6	↔		35	A19/S6	←	
AD9	7	↔		34	BHE/S7	←	
AD8	8	↔		33	MN/MX	←	
AD7	9	↔		32	RD	←	
AD6	10	↔		31	Hold	←	(RQ/GT0)
AD5	11	↔		30	HLD/A	←	(RQ/GT1)
AD4	12	↔		29	WR	←	(LOCK)
AD3	13	↔		28	M/IO	←	(S2)
AD2	14	↔		27	DT/R	←	(S1)
AD1	15	↔		26	DEN	←	(S0)
AD0	16	↔		25	ALE	←	(QS0)
NMI	17	→		24	INTA	←	(QS1)
INTR	18	→		23	TEST	←	
CLK	19	→		22	READY	←	
GND	20	←		21	RESET	←	

MN/MX (PIN 33)

The 8086 microprocessor can work in two modes of operations: Minimum mode and Maximum mode.

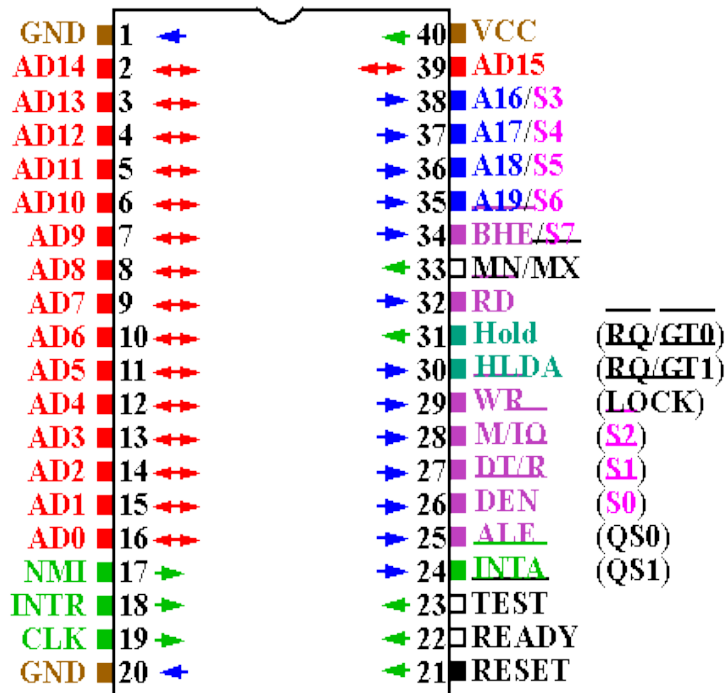
In the **minimum mode** ($MN/\overline{MX} = \text{High}$) microprocessor do not associate with co-processors mode of operation and can not be used for multiprocessor systems.

In the **maximum mode** ($MN/\overline{MX} = \text{Low}$) 8086 can be used in multiprocessor or co-processor configuration

8086/8088 Pin Out

8086 CPU

MIN MODE (MAX MODE)



In Minimum mode pin 33 is tied to VCC. 8086 itself generates all the bus control signals.

DT/R' (Data Transmit/Receive) Output signal from the processor to control the **direction of data** flow through the data transceivers. If **1 transmit if 0 receive**.

DEN (Data Enable) Output signal from the processor used as **to activate the external data bus buffer**.

ALE (Address Latch Enable) Used to **demultiplex the address and data lines** using external latches

M/I $\bar{O}$: Used to differentiate **memory access and I/O access**. For memory reference instructions, it is high. For IN and OUT instructions, it is low.

8086/8088 Pin Out

8086 CPU

MIN MODE (MAX MODE)

GND	1	←	40	VCC
AD14	2	↔	39	AD15
AD13	3	↔	38	A16/S3
AD12	4	↔	37	A17/S4
AD11	5	↔	36	A18/S5
AD10	6	↔	35	A19/S6
AD9	7	↔	34	BHE/S7
AD8	8	↔	33	□ MN/MX
AD7	9	↔	32	RD
AD6	10	↔	31	Hold (RQ/GT0)
AD5	11	↔	30	HLDA (RQ/GT1)
AD4	12	↔	29	WR (LOCK)
AD3	13	↔	28	M/IO (S2)
AD2	14	↔	27	DT/R (S1)
AD1	15	↔	26	DEN (S0)
AD0	16	↔	25	ALE (QS0)
NMI	17	→	24	INTA (QS1)
INTR	18	→	23	□ TEST
CLK	19	→	22	□ READY
GND	20	←	21	RESET

In Minimum mode pin 33 is tied to VCC. 8086 itself generates all the bus control signals.

WR: Write control signal, whenever processor **writes data to memory** or I/O port. Active low.

INTA (Interrupt Acknowledge): When the **interrupt request is accepted** by the processor, the output is low on this line

HOLD Input signal to the processor from the bus masters as a request to grant the control of the bus. Usually used by the **DMA controller** to get the control of the bus.

HLDA (Hold Acknowledge) Acknowledge signal by the processor to the bus master requesting the control of **the bus through HOLD**. The acknowledge is asserted high, when the processor accepts HOLD